# A Reproducible Certificate for the Brass–Sharifi Lower Bound in Lebesgue's Universal Cover Problem

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**Abstract**

Brass and Sharifi proved the lower bound 0.832 for the convex form of Lebesgue's universal cover problem by combining geometric estimates with a computer search over placements of a disk, an equilateral triangle, and a regular pentagon. This paper gives a certificate-based reproduction of that computation. The finite record consists of an adaptive ledger, a terminal-route replay, three local lower-bound certificate families, compact integrity audits for large tables, and a proof-obligation layer connecting the replayed data to the lower-bound statement. Under the specified verifier, acceptance of the finite certificate implies the Brass–Sharifi convex lower bound $\alpha_{\mathrm{cvx}} \geq 0.832$. The certificate concerns only the convex Brass–Sharifi lower bound statement: it claims neither a numerical improvement nor a lower bound for the unrestricted nonconvex problem, and proof-assistant formalization and independent external verification remain outside the present scope.

## 1 Introduction

Lebesgue's universal cover problem asks for a planar set that contains a copy of every planar set of diameter one. In the convex version considered here, the covering set is required to be convex. Thus, a convex set $K \subset \mathbb{R}^2$ is admissible if, for every set $S \subset \mathbb{R}^2$ with $\mathrm{diam}(S) = 1$, there is a rigid motion $g$ such that $g(S) \subset K$. Equivalently, $K$ contains a congruent copy of every diameter-one set. Let

$$\mathcal{U}_{\mathrm{cvx}} = \left\{ K \subset \mathbb{R}^2 : \begin{array}{l} K \text{ is convex, and for every } S \subset \mathbb{R}^2 \text{ with } \mathrm{diam}(S) = 1, \\ \text{some rigid motion } g \text{ satisfies } g(S) \subset K \end{array} \right\}.$$

The convex quantity studied in this paper is

$$\alpha_{\mathrm{cvx}} = \inf_{K \in \mathcal{U}_{\mathrm{cvx}}} \mathrm{area}(K).$$

All lower-bound statements below concern this convex quantity. No claim is made here for the unrestricted nonconvex version of the problem.

The convex problem remains open; see Baez, Bagdasaryan, and Gibbs [2] for background on the broader covering problem. The best known lower bound for the convex problem is the Brass–Sharifi bound 0.832 [1], while the smallest known convex covering construction has area 0.8440935944 in the work of Gibbs [3].

Brass and Sharifi reduced the lower-bound problem to a three-test-set placement problem. They selected a disk, an equilateral triangle, and a regular pentagon, all of diameter one, and estimated the area of their convex hull over a normalized space of relative placements. Their proof combines geometric inequalities with a computer search and establishes the lower bound 0.832.

The original proof was written before the current practice of publishing replayable computational artifacts became common. It describes the estimates used in the search and reports aggregate computation counts. However, it does not present the computation as a replayable certificate in the sense used here. In particular, it does not provide a published terminal-route ledger, a complete replay table, table-level integrity audits for the large certificate data, or an explicit proof-obligation layer connecting the computation to a final verification principle. The purpose of this paper is to make that computation replayable as a finite certificate. The numerical bound itself is unchanged.

We use BS0832 as a label for this reproduction and for the certificate described below. The label is not a theorem number and not a new lower bound; it refers only to the Brass–Sharifi 0.832 computation reconstructed here. The contribution of the present work is the certificate organization: an adaptive ledger and terminal-route replay, three local certificate families, compact integrity audits, explicit proof obligations, and a scope record. In this form the older computation becomes a finite, deterministic, and auditable verification object rather than only an aggregate computational report.

The continuous placement domain is covered by finitely many terminal routes. Each terminal route is assigned a local certificate accepted by the verifier under the stated guard-accounting model. The proof-obligation layer records the finite assertions needed to pass from the route-by-route checks to the global statement. Thus, the lower-bound computation is represented not by a single final decimal, but by a finite cover of the placement domain together with a local lower-bound certificate on every terminal part of that cover.

**Outline.** Section 2 recalls the Brass–Sharifi placement problem and fixes the domains used in the certificate theorem. Section 3 describes the certificate model and verification assumptions. Section 4 describes the adaptive ledger and terminal routes. Section 5 gives the three local certificate families. Section 6 states the adopted Branch-B domain route. Section 7 records the proof-obligation layer. Section 8 gives the certificate theorem and the convex universal-cover consequence. Section 9 records the scope limits of the reproduction. Section 10 concludes. Appendix A lists the verification obligations used in the certificate theorem.

## 2 The Brass–Sharifi placement problem

Let

$$\begin{aligned} C &= \text{the disk of diameter } 1, \\ T &= \text{the equilateral triangle of diameter } 1, \\ P_5 &= \text{the regular pentagon of diameter } 1. \end{aligned}$$

If $K \in \mathcal{U}_{\mathrm{cvx}}$, then $K$ contains congruent copies of $C$, $T$, and $P_5$. Since $K$ is convex, it also contains the convex hull of those three copies. Hence, a uniform lower bound for the area of such convex hulls, over all relative placements, is a lower bound for area$(K)$. The three sets are used as necessary test sets for a lower-bound argument; the paper does not require them to represent all diameter-one sets.

Following the Brass–Sharifi normalization, we fix the disk at the origin and fix the orientation of the triangle. A normalized placement is described by

$$v = (\rho, x_3, y_3, x_5, y_5).$$

We also write

$$u_3 = (x_3, y_3), \qquad u_5 = (x_5, y_5).$$

Here $u_3$ is the triangle translation, $\rho$ is the pentagon rotation, and $u_5$ is the pentagon translation. The symbol $R_\rho$ denotes rotation by angle $\rho$ about the fixed origin. Since the regular pentagon has five-fold rotational symmetry, the pentagon angle is recorded modulo $2\pi/5$ in the certificate's fundamental interval. Define

$$\begin{aligned} H(v) &= \operatorname{conv}\left(C \cup (T + u_3) \cup (R_\rho P_5 + u_5)\right), \\ A(v) &= \operatorname{area}(H(v)). \end{aligned}$$

Figure 1 illustrates this normalized placement and the corresponding hull.

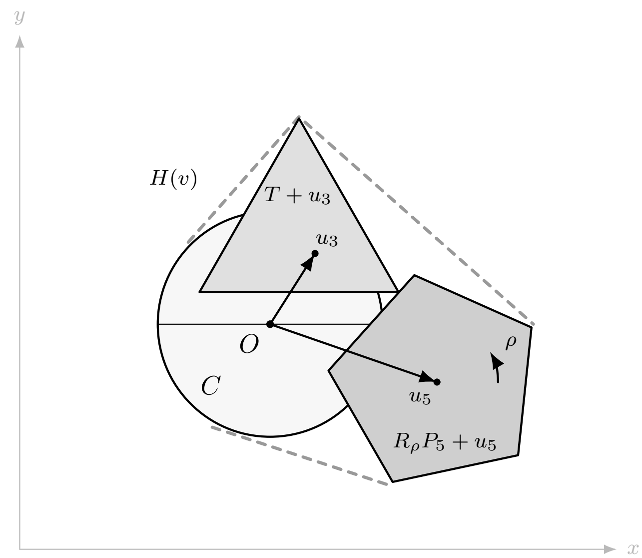


Figure 1: Schematic normalized placement of the three Brass–Sharifi test sets. The disk $C$ is fixed at the origin, the triangle is translated by $u_3 = (x_3, y_3)$, and the pentagon is rotated by $\rho$ and translated by $u_5 = (x_5, y_5)$. The enclosing curve represents $H(v) = \operatorname{conv}(C \cup (T + u_3) \cup (R_\rho P_5 + u_5))$. The drawing is not to scale.

The placement-level lower-bound statement represented by the Brass–Sharifi computation is

$$A(v) \geq 0.832 \qquad \text{for every admissible normalized placement } v.$$

We distinguish three domain objects. Let $\Omega_{\mathrm{adm}}$ be the reduced admissible normalized placement domain represented in the Brass–Sharifi certificate reduction; it is not the unreduced raw translation space. The reduction from the unreduced relative-placement space to $\Omega_{\mathrm{adm}}$ is treated as part of the Brass–Sharifi normalization and the certificate domain model. The present certificate verifies the finite Branch-B replay over this recorded reduced domain; it does not independently rederive the full symbolic reduction from the unreduced translation space. Let $\Omega_B$ be the enlarged parameter domain used in the Branch-B replay. Let $\mathcal{R}$ be the finite set of terminal routes, and let $\Omega_r$ be the subdomain represented by route $r \in \mathcal{R}$. Each terminal-route domain is represented in the certificate by recorded interval bounds on the five placement parameters. The finite replay is applied to these recorded reduced-domain objects, and the domain relation checked by the certificate is

$$\Omega_{\mathrm{adm}} \subseteq \Omega_B \subseteq \bigcup_{r \in \mathcal{R}} \Omega_r.$$

This enlarged-domain route is conservative: proving the inequality on $\Omega_B$ is stronger than proving it only on $\Omega_{\mathrm{adm}}$, because $\Omega_{\mathrm{adm}} \subseteq \Omega_B$. The enlargement concerns the parameter domain used for replay; the test sets $C$, $T$, and $P_5$ remain the fixed diameter-one sets defined above. The Branch-B domain record is the certificate-level object used for the finite replay rather than a closed-form symbolic Branch-A reduction, and its correctness is one of the domain-route obligations of the verification model. For each route $r$, the certificate assigns a local verification object accepted under the stated guard-accounting model and proving

$$A(v) \geq 0.832 \qquad (v \in \Omega_r).$$

The global placement inequality follows once the domain relation and all local inequalities have been verified.

## 3 Certificate model and verification assumptions

The BS0832 certificate is treated in this paper as a finite object together with a deterministic verifier $\mathsf{V}$. The route data describe how the normalized placement domain is subdivided, while the local certificate data discharge the terminal-route domains. The verifier checks the structural consistency of these data before the proof-obligation layer is used for the final aggregation. We use the following terminology.

**Adaptive ledger.**
A parent-child record of the recursive subdivision tree. Each edge records a refinement step in the search; the terminal nodes are the subproblems that must be discharged by local certificates.

**Terminal route.**
A terminal subdomain $\Omega_r$ in the adaptive tree together with the certificate family that discharges it. Terminal routes are the leaves of the replayed certificate.

**Local certificate.**
A finite verification object attached to a terminal route and proving the lower bound on that subdomain under the verification model.

**Directed interval certificate.**
A local certificate family based on directed interval lower bounds. It certifies the largest part of the terminal-route space.

**Local tensor certificate.**
A second local certificate family, organized by tensor members and packages. It supplies package-level lower bounds for terminal routes not discharged by the directed interval family.

$h = 0.004$ **bridge.**
A frozen local bridge certificate at mesh scale $h = 0.004$. It imports a small set of local witness rows into the global replay.

**Branch A and Branch B.**
Branch A denotes a symbolic domain-reduction route that is not used here as a completed proof. Branch B denotes the enlarged-domain replay route adopted by the certificate.

**Guard accounting.**
The explicit subtraction of numerical safety margins before a local lower bound is accepted. The acceptance threshold is the minimum post-guard margin required by the verifier after all recorded guards have been applied.

**Replay audit.**

The check that every terminal route is present, assigned to a certificate family, and free of duplicate or empty accepted-route records.

**Integrity audit.**

A compact record of row counts and block hashes for large certificate tables. Such audits are used to detect data drift without reproducing every table row in the paper.

**Proof obligations.**

A finite list of assertions that connect the replayed certificate data to the final BS0832 claim. The obligations are grouped as OB-A through OB-F.

The verification model is organized around an input–check–output interface. The input is the finite collection of certificate records: the adaptive ledger, the terminal-route replay, the route-to-family assignments, the local-family acceptance records, and the integrity audits. Terminal-route domains are represented in the artifact by recorded interval-domain data on the placement parameters. The verifier $\mathsf{V}$ first checks the structural data: ledger consistency, terminal-route completeness, non-emptiness of recorded route domains, parent-child consistency in the replay, uniqueness of route identifiers, and assignment of each accepted route to exactly one local certificate family. It then checks the numerical and integrity data: row-count records, block-hash records, and the post-guard local acceptance records. Its output is accept or reject. In particular, counts such as "zero violations" are outputs of these replay checks, not independent mathematical assumptions.

The search or generation procedure that produced the certificate is not part of the trusted base. What is intended to be checked is the finite certificate together with the verifier rules and the proof obligations stated in this paper. The accompanying artifact is intended to supply the machine-readable records and verifier with the public release of the manuscript; this paper refers to them abstractly as the BS0832 certificate and the verifier $\mathsf{V}$ and reports only the certificate-level summaries needed for the mathematical implication. Table 1 summarizes these logical components.

Table 1: BS0832 certificate model.

| Logical component | Role in the verification model |
|---|---|
| Certificate records | Contain the adaptive ledger, route-domain records, terminal-route cover, route-to-family assignments, and local certificate data. |
| Verifier | Parses the finite records and checks structural consistency, interval-domain records, route assignments, row counts, post-guard margins, and integrity records. |
| Integrity records | Bind large tables by compact count and block-hash information in the accompanying artifact, preventing unnoticed data drift. |
| Proof-obligation record | Lists the finite checks and aggregation assertions by which the replayed data enter the certificate theorem. |
| Scope record | Records the declared scope of the reproduction and the explicit non-claims under the stated verification model. |

The same verification model is used locally on each terminal route. The three local families described in Section 5 differ in how their post-guard lower-bound records are produced; the common post-guard interface is stated there.

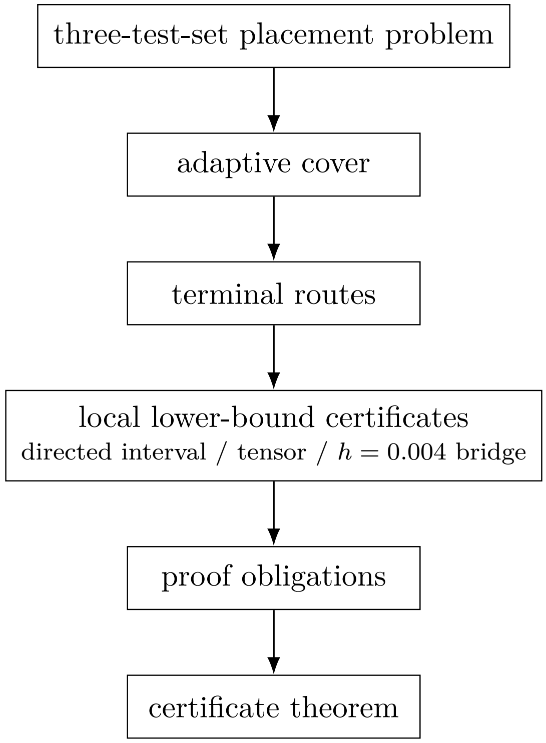


Figure 2: Structure of the BS0832 certificate. The adaptive cover produces terminal routes, local lower-bound certificates discharge those routes, and the proof-obligation layer aggregates the verified local claims into the certificate theorem. Detailed counts are given in Tables 2 and 3.

## 4 Certificate architecture

The adaptive part of the certificate is a finite tree. Its recorded parent-child relation has 379,192 edges, and its leaves give 356,816 terminal routes. Each terminal route is assigned to exactly one of three certificate families. The replay audit verifies that no terminal-route key is duplicated, no accepted route is empty, and no child identifier is duplicated. The numerical distribution of the route families is given in Table 2.

Table 2: Terminal route families.

| Route family | terminal routes |
|---|---:|
| Directed interval certificate | 338,367 |
| Local tensor certificate | 18,380 |
| $h = 0.004$ bridge containment | 69 |
| Total | 356,816 |

Table 3: Certificate component summary.

| Component | rows/items | violations vs. 0.832 | certification role |
|---|---|---|---|
| Adaptive parent-child ledger | 379,192 | – | full route tree |
| Adaptive terminal routes | 356,816 | 0 | route closure |
| Directed interval rows | 41,261 | 0 | main directed route |
| Local tensor members | 8,751 | 0 | tensor route |
| Local tensor packages | 125 | 0 | package-level tensor binding |
| $h = 0.004$ bridge witnesses | 282 | 0 | local proof bridge |

The route counts and supporting data counts have different granularities. A directed row may support several terminal-route acceptances; a tensor package binds many tensor members and can discharge a collection of terminal routes; the $h = 0.004$ bridge imports witness rows for a small residual route set. The aggregation layer therefore tracks both terminal-route coverage and the supporting certificate tables.

# 5 Local certificate families

For a terminal route $r$, a local verifier supplies a post-guard lower-bound record. Abstractly, the record gives a certified value $L_r^{\text{post}}$ such that, under the stated numerical-kernel assumptions,

$$L_r^{\text{post}} \leq \inf_{w \in \Omega_r} A(w), \qquad L_r^{\text{post}} - 0.832 \geq 10^{-7},$$

where $10^{-7}$ is the verifier's acceptance threshold. Thus, $L_r^{\text{post}}$ is a certified lower estimate for all values of $A$ on $\Omega_r$ after the recorded numerical guard has been subtracted. The guard is a conservative allowance for enclosure and rounding effects in the local lower-bound computation. Hence, for every $v \in \Omega_r$,

$$A(v) \geq \inf_{w \in \Omega_r} A(w) \geq L_r^{\text{post}} \geq 0.832 + 10^{-7} > 0.832.$$

Thus, an accepted local record certifies $A(v) \geq 0.832$ throughout the route domain $\Omega_r$. The final aggregation uses only this route-level inequality and the route assignment. The directed interval, tensor, and bridge families differ in how they produce the post-guard records, not in how their accepted records enter the aggregation. No proof-assistant formalization of the numerical kernels is claimed.

The three local families partition the accepted terminal routes in the replay: directed interval routes, local tensor routes, and the $h = 0.004$ bridge routes.

## 5.1 Directed interval certificates

The directed interval family is the primary component of the replay, discharging 338,367 terminal routes and using 41,261 directed rows. These records encode interval lower-bound checks for grouped terminal routes after the recorded numerical guards have been applied. No route in this family violates the 0.832 bound under verifier inspection. After guard accounting, the smallest directed margin reported by the certificate is approximately

$$4.307276422 \cdot 10^{-6}.$$

Since this is above the acceptance threshold $10^{-7}$, the directed family is accepted by the verifier under the stated guard-accounting model on its assigned terminal routes.

### 5.2 Local tensor certificates

The local tensor family handles 18,380 terminal routes through package-level lower-bound records. It is backed by 8,751 tensor members organized into 125 packages, with member-to-package consistency checked before acceptance. The tensor replay likewise contains no verified violation of the 0.832 bound, and the smallest tensor margin after guard accounting is approximately

$$2.318262102 \cdot 10^{-5}.$$

This value is also above the acceptance threshold. The tensor family has its own structural checks: member-to-package binding, absence of duplicate members, absence of package-member mismatches, and consistency with the terminal-route replay.

Table 4: Post-guard margins.

| Family | Minimum post-guard margin above 0.832 |
|---|---|
| Directed interval | $4.307276422 \cdot 10^{-6}$ |
| Local tensor | $2.318262102 \cdot 10^{-5}$ |
| Verifier acceptance threshold | $10^{-7}$ |

### 5.3 The $h = 0.004$ bridge

The $h = 0.004$ bridge covers the residual set of 69 terminal routes. It imports a frozen mesh-scale local witness table with 282 rows for routes not handled by the two larger families. The bridge witnesses pass the 0.832 bound check without violation. In the final aggregation this component is treated as a separate lemma rather than being absorbed into the directed or tensor families.

## 6 Domain route

A subtle point in the reproduction is the domain reduction. The certificate does not assert that the symbolic Branch-A domain reduction has been closed. The adopted route is instead the Branch-B enlarged-domain replay. This replay records the relation

$$\Omega_{\mathrm{adm}} \subseteq \Omega_B \subseteq \bigcup_{r \in \mathcal{R}} \Omega_r$$

and binds the terminal routes to the three certificate families. The finite replay is applied to the reduced admissible domain recorded by the certificate, not to the unreduced raw translation space. Under the certificate model, the Branch-B domain records certify that the recorded terminal-route domains cover $\Omega_B$; together with the recorded inclusion $\Omega_{\mathrm{adm}} \subseteq \Omega_B$, this gives the finite cover used below.

Table 5: Domain-route status.

| Route | Status | Role in this paper |
|---|---|---|
| Branch A | not claimed closed | not used as a completed proof route |
| Branch B | adopted | supplies the enlarged-domain replay and terminal-route cover |

All subsequent claims in this paper use Branch B, not Branch A. The enlarged-domain inclusion is a one-way strengthening: because $\Omega_{\mathrm{adm}} \subseteq \Omega_B$, a lower bound verified throughout $\Omega_B$ applies immediately to all admissible normalized placements. The certificate uses this recorded Branch-B domain relation rather than a closed-form symbolic Branch-A reduction. This distinction is part of the verification model and is also recorded in the proof-obligation layer.

# 7 Proof-obligation layer

The proof-obligation layer records how the Branch-B domain route, terminal-route replay, local certificate families, and final aggregation enter the certificate theorem.

The last layer of the certificate is a finite list of nineteen proof obligations. These obligations identify the checks and mathematical assertions needed by the adaptive ledger, the directed interval family, the local tensor family, the $h = 0.004$ bridge, the domain route, and the final aggregation. Table 6 summarizes the six groups. The complete list is given in Appendix A.

Table 6: Proof-obligation groups.

| Group | Component | Core requirement |
|---|---|---|
| OB-A | adaptive ledger | terminal routes have a closed parent-child ledger |
| OB-B | directed interval family | directed records meet the post-guard margin requirement |
| OB-C | local tensor family | tensor members bind to packages without gaps |
| OB-D | $h = 0.004$ bridge | bridge witnesses are imported and checked |
| OB-E | domain route | Branch B is adopted; Branch A is not claimed closed |
| OB-F | final aggregation | the final statement is exactly the BS0832 bound |

The obligation layer separates the replay of numerical and combinatorial data from the mathematical assertions used in the final aggregation. In particular, the obligations are phrased as record checks, verifier checks, or explicitly stated aggregation assertions; they are not intended to replace local verification by merely assuming the desired conclusion.

# 8 Certificate theorem and convex universal-cover consequence

We now separate the purely finite-cover implication from the certificate-level assertion that the verifier and proof obligations supply the hypotheses of that implication. Figure 3 summarizes this dependency structure.

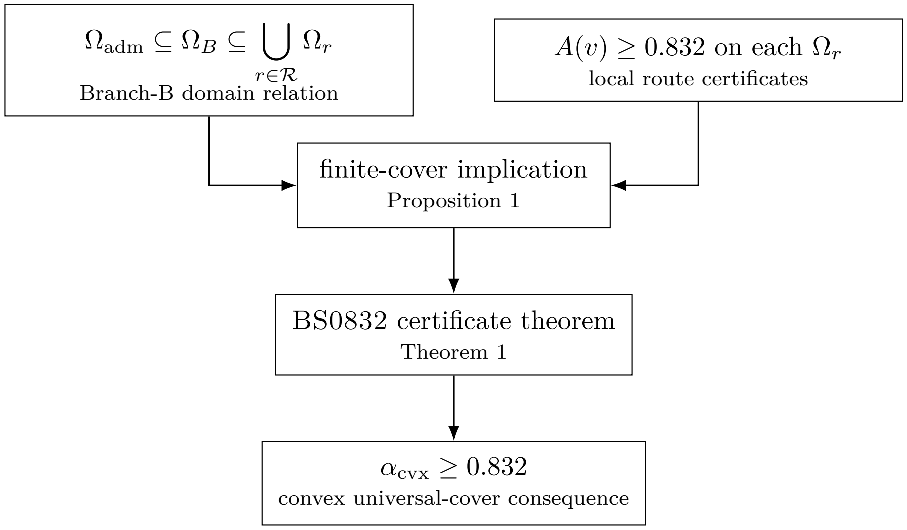


Figure 3: Proof dependency structure. The Branch-B domain relation and the local route certificates provide the hypotheses of the finite-cover implication. The certificate theorem applies this implication under the specified verifier, and the convex universal-cover consequence follows from the Brass–Sharifi normalization and the recorded admissible-domain reduction.

**Proposition 1** (Finite-cover implication)**.** *Assume that*

$$\Omega_{\mathrm{adm}} \subseteq \Omega_B \subseteq \bigcup_{r\in\mathcal{R}} \Omega_r,$$

*and that for every $r \in \mathcal{R}$ the assigned local certificate proves*

$$A(v) \geq 0.832 \qquad (v \in \Omega_r).$$

*Then $A(v) \geq 0.832$ for every admissible normalized placement $v \in \Omega_{\mathrm{adm}}$.*

*Proof.* Let $v \in \Omega_{\mathrm{adm}}$. By the domain inclusion, $v \in \Omega_B$, and then $v$ lies in at least one terminal-route domain $\Omega_r$. The local certificate assigned to that route gives $A(v) \geq 0.832$ on $\Omega_r$. Since $v$ was arbitrary, the inequality holds on $\Omega_{\mathrm{adm}}$. □

**Theorem 1** (BS0832 certificate theorem, under the stated verification model)**.** *Let the* BS0832 *certificate be the finite certificate consisting of the adaptive ledger, terminal-route cover, local certificate records, integrity records, and proof-obligation record described above. If the verifier* $\mathsf{V}$ *accepts this certificate and the obligations* OB-A–OB-F *are discharged under the verification model of Section 3, then*

$$A(v) \geq 0.832$$

*for every admissible normalized placement $v \in \Omega_{\mathrm{adm}}$.*

*Proof.* Verifier acceptance and the obligations OB-A–OB-F supply the replayed route data, the Branch-B domain relation, the route-to-family assignments, and the accepted local post-guard records. By the local interface of Section 5, those accepted local records give the route-level inequalities required in Proposition 1. The placement-level inequality follows from that proposition. □

**Corollary 1** (Convex universal-cover consequence)**.** *Under the hypotheses of Theorem 1, every convex universal cover has area at least* 0.832*. Consequently $\alpha_{\mathrm{cvx}} \geq 0.832$.*

*Proof.* Let $K \in \mathcal{U}_{\mathrm{cvx}}$. Then $K$ contains congruent copies of $C$, $T$, and $P_5$. Since $K$ is convex, it contains the convex hull of those three copies. After applying the Brass–Sharifi normalization and the recorded admissible-domain reduction, these copies correspond to an admissible placement $v \in \Omega_{\mathrm{adm}}$. Area is invariant under the normalization, and the corresponding normalized hull has area $A(v)$. By Theorem 1, this convex hull has area at least 0.832. Hence $\mathrm{area}(K) \geq 0.832$. Taking the infimum over $K \in \mathcal{U}_{\mathrm{cvx}}$ gives $\alpha_{\mathrm{cvx}} \geq 0.832$. □

## 9 Verification scope and limitations

This paper describes a certificate theorem for the existing Brass–Sharifi lower-bound computation in the convex setting. It does not claim a numerical improvement of the Brass–Sharifi bound, a nonconvex universal-cover lower bound, a proof-assistant formalization (such as a Lean, Coq, or Isabelle development), a completed Branch-A symbolic reduction, or independent external verification. The listed obligations have been reviewed by the author under the stated verification model; this review records acceptance of the certificate assumptions, not an independent external verification. The accompanying machine-readable certificate records and verifier are intended to be distributed as supplementary artifacts with the public release of this manuscript; the present paper reports the summaries needed for the certificate implication rather than a full artifact manifest.

## 10 Conclusion

We have described a reproducible finite record for the Brass–Sharifi lower bound in the convex version of Lebesgue's universal cover problem. The numerical bound is unchanged, but the record makes explicit the finite route structure, local lower-bound checks, guard accounting, and proof obligations supporting the statement $A(v) \geq 0.832$ on the normalized admissible placement domain. Under the stated verification model, these ingredients imply the convex universal-cover consequence $\alpha_{\mathrm{cvx}} \geq 0.832$. In this sense, the original Brass–Sharifi computation is placed in a form that can be replayed and audited independently of the original search program. The same finite-record structure could also serve as a basis for later independent review or proof-assistant formalization.


## Acknowledgements

The author used ChatGPT as an auxiliary tool for organizing implementation notes, checking reproducibility workflows, and improving exposition. The mathematical claims, computations, certificate judgments, and final text are the responsibility of the author.


## References


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# A Verification obligations

The proof-obligation layer used in the certificate theorem consists of the following nineteen obligations. The labels are stable identifiers for the checks and assertions; they are not new mathematical constants.

Table A.1: Verification obligations.

| Label | Component | Verification check or aggregation assertion |
|---|---|---|
| OB-A-001 | adaptive ledger | the verifier parses the parent-child ledger and rejects unless it has the recorded recursive subdivision structure |
| OB-A-002 | adaptive ledger | the verifier rejects unless terminal-route records are present and complete for replay |
| OB-A-003 | adaptive ledger | the verifier rejects unless route keys and child identifiers pass duplicate, emptiness, and consistency checks |
| OB-B-001 | directed interval | the verifier rejects unless directed interval records satisfy the input schema and hypotheses required by the directed-bound verifier |
| OB-B-002 | directed interval | the verifier rejects unless directed accepted-route records are bound to terminal-route records |
| OB-B-003 | directed interval | for directed records, the verifier rejects unless post-guard margins exceed the acceptance threshold for the 0.832 bound |
| OB-B-004 | directed interval | the verifier rejects unless the integrity audit for the directed interval records is accepted |
| OB-B-005 | directed interval | the verifier computes the directed-family violation counter from checked records and rejects unless it is zero for the 0.832 bound |
| OB-C-001 | local tensor | the verifier rejects unless tensor member and package records satisfy the input schema and hypotheses required by the tensor verifier |
| OB-C-002 | local tensor | the verifier rejects unless tensor member-to-package binding has no recorded gaps, duplicates, or mismatches |
| OB-C-003 | local tensor | the verifier computes tensor package checks and rejects unless the bound-violation counter is zero |
| OB-D-001 | $h = 0.004$ bridge | the verifier imports bridge witness rows, binds them to residual routes, and rejects unless the bound-violation counter is zero |
| OB-E-001 | domain route | Branch B is the adopted domain route for this certificate |
| OB-E-002 | domain route | Branch A is not claimed as a completed symbolic proof |
| OB-E-003 | domain route | the verifier rejects unless the Branch-B domain records certify, under the certificate domain model, that the recorded terminal-route domains cover the adopted enlarged domain $\Omega_B$ and hence the admissible domain $\Omega_{\mathrm{adm}}$ |
| OB-F-001 | final aggregation | the final statement is exactly the BS0832 0.832 reproduced lower-bound statement |
| OB-F-002 | final aggregation | no stronger numerical lower-bound claim than the BS0832 0.832 bound is included |
| OB-F-003 | final aggregation | proof-assistant formalization and independent external verification remain outside the present scope |
| OB-F-004 | final aggregation | accepted route, local-family, and domain obligations aggregate through the finite-cover implication to the certificate theorem |